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A review of the specificity of exercises designed for conditioning the lumbar extensors

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ABSTRACT

Objective To review the specificity of exercises designed to condition the lumbar extensor musculature (ie, lumbar erector spinae and multifidus).

Methods A review of studies examining effects of exercises designed to condition the lumbar extensors was conducted. Included were studies that examined the acute activation and chronic adaptation of the lumbar extensor musculature in response to benches and roman chair trunk extensions, free weights exercises (ie, deadlifts, squats, good-mornings, etc), floor and stability ball exercise (ie, trunk extensions, bridging, four-point kneeling, etc) and resistance machines (ie, those with and without pelvic restraints).

Results Evidence suggests that the reviewed exercises designed to condition the lumbar extensors all may result in significant activation of this musculature during their performance. However, examination of training studies shows that for benches and roman chair trunk extensions, free weights exercises, floor and stability ball exercise and resistance machines without appropriate pelvic restraints, evidence suggests that they may be less effective for inducing chronic adaptations in the lumbar extensors as a result of their performance. Contrastingly, resistance machines that employ appropriate pelvic restraint to isolate lumbar extension are better evidenced to confer specific adaptations to the lumbar extensors.

Conclusions Numerous exercise approaches have been designed with the intention of conditioning the lumbar extensors. Those examined appear to activate the lumbar extensors; however, the specificity of many of these exercises for producing chronic adaptations may be questionable, potentially due to the compound nature of them allowing involvement of other musculature such as the hip extensors. Many of the reviewed exercises offer potential to condition the lumbar extensors, however, isolation of lumbar extension through appropriate pelvic restraint appears important for optimising specific adaptations in the lumbar extensors.

INTRODUCTION

For athletes as well as the general population alike, conditioning the muscles of the lower back could be considered an important aspect of physical conditioning. Low back pain (LBP) is a prevalent condition among the general populace¹⁻⁶ and athletes alike from various sports.⁷⁻¹¹ It has been suggested that deconditioning of the lumbar extensor musculature (lumbar erector spinae and multifidus) is a risk factor for low back injury and pain, and that conditioning them through exercise might offer an effective means of reducing this risk.¹²⁻¹⁵

Early efforts at conditioning the lumbar extensors were reported upon by DeLorme and Watkins.^{16 17} They reported at this time on specialised equipment,

designed and intended to provide exercise specifically to the lumbar extensors, by attempting to restrain concurrent pelvic movement. They noted that, with increasing strength, symptoms of LBP were relieved. Studies in occupational settings specifically strengthening the lumbar extensors of employees through preventative exercise have also shown reduced low back injury risk.^{18 19} Exercise that conditions and strengthens is suggested as valuable in effectively 'prehabilitating' and reducing injury risk for athletes.²⁰ Thus, specifically prehabilitating the lumbar extensors to reduce low back injury and LBP risk might be deemed an important goal of training and identifying the most effective means an important topic to pursue.

However, the lumbar extensor musculature has been suggested to be notoriously difficult to condition.¹²⁻¹⁵ It is further suggested that valid testing of strength or endurance of the lumbar extensors requires isolation of the lumbar spine through restraint of the pelvis.²¹⁻²⁹ Otherwise, due to the longer moment arms and relatively larger cross-sectional areas (CSA), the hip extensors (gluteus and hamstrings) may contribute a greater degree of measured torque.³⁰ This may also influence the effectiveness of exercises designed for the lumbar extensors as during compound exercise the larger musculature is suggested to take over the load.³¹

Progressive resistance exercise used historically in treating musculoskeletal disorders such as LBP, as well as the suggested difficulty in conditioning the lumbar extensors due to the influence of the hip extensors, has resulted in more specific approaches and devices being developed in order to condition the lumbar extensors. Isolation of the lumbar extensors is potentially important in valid testing.²¹⁻²⁹ However, this does not automatically imply exercises that do not isolate are ineffective in conditioning them specifically.

Exercise approaches to conditioning the lumbar extensors have been previously reviewed with regards to treating LBP.^{14 15 32} However, discussion of the specificity of exercise for conditioning the lumbar extensors was absent from earlier reviews. Mayer *et al*³³ offered the tentative suggestion that isolated lumbar extension (ILEX) exercise using dynamometers appears to be the best option for treating LBP. Though, their conclusion may not be applicable to the consideration of prehabilitation. Thus this present review offers a discussion and comparison of the specificity of exercise approaches aimed at conditioning the lumbar extensors in asymptomatic populations for prehabilitation purposes. The specificity of exercises in acutely activating the lumbar extensors in addition to their specificity in producing chronic adaptations will be examined.

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METHODS

Previous reviews' reference lists were searched in addition to SPORTDiscus, PubMed and Google Scholar databases (utilising search terms including combinations of 'lumbar extension', 'lumbar exercise', 'lumbar strength', 'lumbar endurance', 'lumbar spine', 'low back exercise', 'lumbar activation' and 'lumbar EMG' as well as other associated terms, synonyms and combinations of the terms) considering all publications listed up to September 2012. Exercise approaches designed to target the lumbar extensors (lumbar erector spinae and multifidus) were defined (see below) and only studies utilising these and providing testing that could isolate and identify the effect of exercise on the lumbar extensors (also defined below) were included. Thus the results of this review are split into two main sections; (1) effects during performance of exercises upon acute activation, measured using electromyography (EMG), of the lumbar extensor musculature and (2) training studies reporting chronic adaptation in the lumbar extensors as a result of different exercises. These sections will seek to critically compare the specificity of the exercise approaches examined.

Exercises designed for the lumbar extensors

One of the key considerations highlighted by Mayer *et al*³³ is the importance of the actual exercise movement performed to condition the lumbar extensors. Mayer *et al*³³ highlight and define four main types of exercise that are purported to target the lumbar extensor musculature including:

- ▶ Benches and roman chair trunk extensions (TEX)
- ▶ Free weights (ie, deadlifts, squats, good mornings, etc)
- ▶ Floor and stability ball exercise (ie, TEX, bridging, four-point kneeling etc)
- ▶ Resistance machines (ie, those with and without pelvic restraints)

These exercise approaches are defined here as being designed to condition the lumbar extensors. Of these approaches some allow compound TEX to occur (benches and roman chairs, free weights, floor and stability balls and some resistance machines) whereas others provide sufficient pelvic restraints (those providing ILEX—described below). Here each of these purportedly 'specific' exercises will be considered and both their effect on acute lumbar extensor recruitment during performance of the exercise and development of lumbar extensor strength, endurance or hypertrophy through training.

Validity of outcome measures

Although imaging study perhaps provides the most valid means of identifying whether adaptations have occurred in specific musculature as a result of a training intervention, relatively few studies examining the specificity of lumbar extensor exercises have utilised such outcome measures.^{34–36} Another means of differentiating the lumbar extensor musculature from, for example, the hip extension musculature, might be to examine changes in fatigability utilising EMG placed over the lumbar extensors during exercise testing. Some studies have utilised such outcome measures.³⁷ Cross talk is a particular issue when differentiating between the specific lumbar extensor musculature (ie, lumbar erector spinae and multifidus) and placements,³⁸ however, lumbar electrode placements should be easily distinguishable from hip extensor placements and also from thoracic extensor electrode placements and might be considered to potentially represent the lumbar extensor musculature collectively.

Perhaps explaining the relative paucity of studies using the above mentioned methods, it is often common place to determine whether any adaptation has occurred in musculature through use of strength or endurance testing. When examining the lumbar extensors though, consideration should be given to

the tests utilised. Testing of compound movements (ie, TEX) may demonstrate improvement in strength or endurance for a specific movement; but not whether one particular muscle group has improved.²² Isolated testing (ie, ILEX) instead allows specific induction as to whether the musculature under question has improved.²² Due to the importance of differentiating between such modes of testing for examining the specificity of lumbar extensor exercises they are now explained in advance here.

Because of the potential involvement of the hip extensors in TEX²⁹ the pelvis should be restrained and stabilised appropriately for the purpose of testing ILEX.^{16 17 22} If the pelvis is not stabilised then during testing it is difficult to determine the actual source of measured extension torque. Measured torque as a result of TEX may be overstated by the contribution of the hip extensors due to the longer moment arms over which the gluteus and hamstrings exert force and their relatively larger CSA.³⁰ If the pelvis is able to move then measurements of strength obtained may not be valid reflections of the strength of the lumbar extensor musculature.

In addition, the hip extensor musculature has the potential to influence tests of TEX endurance. Measures of endurance using the Biering-Sorenson TEX test show inconsistent results regarding reproducibility.^{39–43} Indeed Kankaanpää *et al*⁴⁴ reported that the hip extensors show significant fatigue indicative of load sharing during the Biering-Sorensen isometric endurance TEX test. Clark *et al*^{45 46} have further quantified this suggesting derecruitment of the lumbar extensors and further recruitment of the hip extensors suggesting they are 'taking over' the load. Contrastingly, those that have examined ILEX endurance using appropriate restraint systems for the pelvis (as described below) indicate consistently reproducible results, perhaps due to the control of the specific musculature used during testing.^{47 48}

This involvement of the hip extensor musculature is not just evidenced in prone TEX tests but also in some seated extension test devices (some of which are purported to provide ILEX) where the participant is seated with knees at 90° and the equipment does not restrain the pelvis posteriorly.⁴⁹ The position of the knees might be important in driving posterior stabilisation of the pelvis against a posterior restraint to specifically isolate the contribution of the lumbar extensors with research suggesting a semiseated position may be optimal by limiting involvement of the hip as well as the thoracic extensors^{27–29} and posterior pelvic stabilisation also appearing essential.^{22 23} Indeed testing with a 90° angle for knee position is shown to overstate torque measures compared with testing that utilises a semiseated position.²⁶

Some manufacturers produce devices that provide the restraint set-up that research suggests is optimal in producing the conditions to test or train ILEX (eg, Lumbar Extension Machine, MedX, Ocala, Florida; BackUp Dynamometer, Priority One Equipment, Grand Junction, Colorado; Lower Back Revival System, OriGENE Concepts BV, Delft, the Netherlands etc.^{47 48 50 51}). Other researchers have also utilised customised seats providing such optimal restraints to use with generic dynamometers^{28 29 52–54} suggesting this is also an option available to researchers and clinicians. Force applied by a footboard to the bottom of the feet is transmitted to the distal end of the femurs above the knees through the lower legs. Forces are then transferred in two directions; driving the femurs towards the rear thus fixing them into the pelvic socket and force pushing them upwards into knee restraints. A thigh restraint can act as a fulcrum redirecting upwards force at the knees to a downwards force on the pelvic sockets. The results are forces that push the pelvis back and downwards thus preventing any rotation, upward or forward movement.²¹ If a posterior pelvic restraint which is free to move about

its own axis is used then, during consequent flexion and extension, movement of the pelvis is evidenced through rotation of the posterior pelvic restraint.¹⁵ This restraint set-up ensures isolation of lumbar extension movement from hip extension movement and thus differentiation of the muscles producing each respectively. However, it should be noted in advance that the movement produced during ILEX is a product of the action of the lumbar extensor musculature (lumbar erector spinae and multifidus) as well as the thoracic portions of the extensor musculature in addition to bilateral contraction of the quadratus lumborum. The relevance of this issue is expanded on within the discussion.

Isolation of the lumbar extensor musculature through restraining both the legs and pelvis posteriorly has been shown to significantly reduce pelvic movement when compared to less rigorous restraints^{22–23} and the restraints described are reported to allow at most only a 3° pelvic rotation,²⁵ likely as a result of unavoidable soft tissue compliance. As noted by Smidt *et al*²² although ultimately the use of compound as well as isolated testing is useful in identifying ‘weak links,’ in the pursuit of an answer regarding the specificity of exercises designed to condition the lumbar extensors a focus should be placed on isolated testing for chronic outcomes.

RESULTS

Acute activation of the lumbar extensors through specific exercise

As noted, for valid testing, isolation of lumbar extension requires adequate restraint of the pelvis in order to limit the involvement of the hip extensors.^{21–29} However, although this is suggested, it does not provide evidence that the same requirement must be present in order to effectively condition the lumbar extensors. Some exercises do provide such isolation allowing ILEX to be performed and have been argued to be potentially the most effective means of conditioning the lumbar extensors as a result of a potentially greater activation of the lumbar extensor musculature.^{33–51} Thus, it is interesting to examine lumbar extensor activation during other exercises as this may offer insight into the most effective means of training them. Each of the noted types of exercises that are performed with the aim of conditioning the lumbar extensors have been shown to activate these muscles when measured using EMG placements over the lumbar extensors. The results however are quite contrasting within themselves as well as in corroboration with data from chronic training studies, as will be discussed.

Roman chair or bench TEX exercise produces a significant activation of the lumbar extensors,^{55–61} which increases further when internal hip rotation and lumbar extension are accentuated during the movement; this being suggested to better isolate the lumbar extensors involvement.⁵⁸ Indeed, even early analysis of lumbar extensor activation during exercise suggested that hyperextension from the prone position produced greatest activation.⁶² Conversely, it has also been shown that when a form of pelvic restraint is used during roman chair TEX, lumbar extensor activation is significantly less than when unrestrained.⁶³ This occurred despite participants’ subjective perceptions of muscular fatigue being more significantly localised to the lumbar extensors when restraints were used. Arguably the EMG results are more objective yet it is curious to find this discrepancy. One study has also compared the activation of the lumbar extensors during roman chair TEX to ILEX finding greater activation at lighter loads for the roman chair and at heavier loads no difference.⁵⁵ More recently Lariviere *et al*⁶¹ reported greater activation and fatigue of the hip extensors compared to the lumbar extensors during roman chair TEX, suggesting this was due to a greater

freedom to change kinematics, which they also reported and hence load shared between the lumbar and hip extensors. Clark *et al*^{45–46} have supported this apparent load sharing and derecruitment of the lumbar extensors during TEX demonstrating that, after a certain relative degree of fatigue had been achieved during TEX (~55%), both between further sets of exercise not to failure⁴⁵ and within a single set of exercise performed to momentary muscular failure,⁴⁶ the lumbar extensors’ recruitment decreases. The hip extensors however show evidence of further recruitment and fatigue suggesting they are ‘taking over’ the load.

Studies have also examined free weight exercises. They have shown exercises such as deadlifts^{64–67} and squats⁶⁵ also produce significant activation of the lumbar extensors with comparative studies suggesting deadlifts to offer the greatest degree of activation.^{65–67} Floor and stability ball-based exercises (commonly referred to as ‘stability’ exercises) aimed at conditioning the lumbar extensors also show activation of the targeted musculature, however there is considerable variability between the individual exercises employed using these approaches and even between studies.^{56–59–68–75} However, some degree of activation of the lumbar extensor musculature through these specific exercise approaches appears to occur and suggests a potential for them to condition them.

The lumbar extensors are also activated when using ‘lower back’ resistance machines, although there is conflicting evidence as to whether this is greater with or without pelvic restraints and hence TEX or ILEX machines. Udermann *et al*⁷⁶ compared lumbar extensor activation during exercise on a machine providing ILEX when the restraints are equipped, with and without the use of the restraints, finding a trend towards greater activation in the restrained condition which approached significance ($p=0.06$). This, however, suggests that the lumbar extensors are, at the least, active in both conditions though perhaps more during ILEX. Walsworth,⁷⁷ this time comparing ILEX with another lower back machine that utilised a less sophisticated restraint, reported no significant differences in lumbar extensors activation between the two machines. Two other studies however have suggested the use of an appropriate pelvic restraint system does result in significantly greater activation of the lumbar extensors.^{27–28} San Juan *et al*²⁷ and Da Silva *et al*²⁸ reported significantly greater activity in the lumbar extensors when appropriate pelvic restraints are utilised. Both these studies utilised relatively low loads during exercise (50% and 40% of maximal voluntary contraction (MVC), respectively) whereas studies by Udermann *et al*⁷⁶ and Walsworth⁷⁷ that have suggested there to be no significant difference in activation of the lumbar extensors utilised greater relative loads (~80% of MVC). Significant lumbar extensor activation observed in an unrestrained yet heavy TEX based movement may be due to the multifidus’ role in attempting to stabilise the pelvis due to its origin at the sacrum⁷⁸ and this may also explain the findings of Kearns *et al*⁵⁵ with regards to roman chair extensions showing no difference compared to ILEX at high loads. Another study has shown a resistance machine that fastens only the knees and feet at a 90° angle does allow significant activation of the lumbar extensor musculature when using a load of 50% MVC, however, the gluteus maximus may fatigue to a significantly greater degree⁴⁹ suggesting that the gluteus maximus may be contributing to the work performed to a greater degree. Contrastingly, Lariviere *et al*²⁹ showed that when a semiseated position and restraints providing ILEX are used there is little activation and fatigue in the hip extensors and that activation and fatigue appears specifically limited to the lumbar extensors.

Despite numerous studies demonstrating that lumbar extensors are active during performance of a variety of different exercises it should be noted this does not provide direct evidence for adaptations as a result of chronic training; it merely suggests the potential. Indeed some studies demonstrate higher levels of lumbar extensor activation than others depending on the exercise, the manner in which it is performed, the load and the presence or absence of pelvic restraints. There is also the potential consideration of electrode placements and which of the lumbar extensor musculature (ie, lumbar erector spinae, multifidus) is being most reflected as it is quite possible that different adaptations could occur in different extensor muscles as a result of different exercises. The involvement of the lower thoracic portion of the extensor musculature is also an important consideration as electrode placements have included those situated on the lower thoracic portion of the extensors (ie, T9, T10, T11 and T12^{28 29 56 61 64 66}), and those ranging across the entirety of the lumbar portion (L1–S1^{27 28 44–46 49 55–77}) with authors referring to many markers interchangeably as being representative of many of the muscles noted. Goncalves and Barbosa⁶⁰ reported that there may be differences in fatigue during TEX between electrodes placed over L4/L5 (referred to as the multifidus) and those placed at L2/L3 (referred to as the iliocostalis lumborum) as have others across different placements, higher as well as lower, with a range of exercise modes.^{28 61 65 66 76} It should again be noted that there is considerable difficulty in avoiding crosstalk between different electrode locations when examining the lumbar extensors³⁸ and perhaps this is why other studies have shown that there is little difference between electrode placements in the lumbar region.^{44 56 64 69 71 74} Yet in comparing lumbar and lower thoracic extensor activity Arokoski *et al*⁵⁶ reported TEX type exercise to produce significant activation in lumbar as well as lower thoracic extensors but offered no comparison between placements which appeared similar. However, during fatiguing TEX exercise it would appear that, in tandem with lumbar extensor derecruitment and further hip extensor recruitment,^{45 46} there is increased recruitment of the lower thoracic extensors indicating further load modulation between the lumbar and lower thoracic extensors perhaps due to greater freedom of kinematics.⁶¹ For free weight deadlift exercises though there appears to be no difference between lumbar or lower thoracic activation.^{64 66 67} Arokoski *et al*⁵⁶ also examined stability exercise but again did not report comparisons between placements. During ILEX exercise it seems there is a greater lumbar extensor activity and fatigue compared with the lower thoracic extensors.^{28 29} Nevertheless, though comparisons within studies and exercises can be offered in this regard, the range of methodologies used make comparison between studies and exercises difficult.

In addition we must also consider that many trunk muscles demonstrate a potential error in amplitude measurements of ~15%.⁷⁹ Although different treatments of EMG signal data render difficulty in comparing between studies, the activation levels measured for all of the exercises examined ranged from <5% to ~150% for values normalised to MVC, with between exercise mode comparisons in studies showing differences ranging from ~3% to ~80%. Evidently there are some comparative studies showing clearly significantly greater activation of the lumbar extensors utilising some approaches over others (ie, greater than 15% difference for bench or ball-based TEX compared to other floor/ball-based exercises,^{56 59} deadlift compared to other free weight exercises,^{65 67} hip extension or TEX based floor exercises compared to other floor exercises^{70–72} and ILEX compared to machine based TEX^{27 28}), however this is variable

between other comparative studies of similar exercises.^{63 69 74 75} Thus, it is not particularly clear whether all the differences noted can arguably be used to prescribe exercises for conditioning the lumbar extensors. It appears unclear whether some differences in activation reported by studies are indeed present or may be due to measurement errors. Further, the normalised values in studies using TEX based movements for determination of MVC may be of particular concern. During TEX based movement the hip extensors likely contribute significantly more torque during determination of MVC^{21–31} and thus EMG recordings for lumbar extensor MVC could be understated when determined this way, that is, may not be maximal themselves. As such this would affect the relative activation measured in the lumbar extensors during TEX exercise, potentially artificially decreasing the measured activation levels in studies that have used TEX based MVCs.^{45 46 56 58 59 63–67 69–72 74 75}

To possibly account for this some studies have utilised analysis of EMG fatigue indices (ie, EMG power spectrum analysis) which allow examination independent of MVC.⁸⁰ These studies do show that exercises such as TEX and unrestrained seated TEX machines show significant fatigue occurs in the lumbar extensors, but that it also occurs in the hip extensors^{44 46 47 49 60 61} and potentially lower thoracic extensors.⁶¹ Training to momentary muscular failure is supported as an optimal stimulus for muscular conditioning^{81–83} as this allows maximal voluntary recruitment of targeted musculature (~85–100%⁸⁴). Despite load sharing of the hip extensors^{45 46} and lower thoracic extensors⁶¹ and derecruitment of the lumbar extensors during TEX to failure, in some exercises they may achieve around >85% of maximal voluntary activation (particularly TEX,⁵⁹ deadlifts and squats,^{65–67} hip extension or TEX based floor exercises^{71 72} and TEX as well as ILEX based extension machines^{27 77}) and a significant degree of fatigue^{44 46 47 49 60 61} which suggests the potential for adaptations may indeed exist through these exercises.

Despite potential methodological differences, the contrasting activation provided by different specific exercises may impact on lumbar extensor conditioning as a result of training using them and thus the value they present in prehabilitation. The degree of activation seen in the lumbar extensors in these studies provides only an indication as to the effectiveness of the exercise for effective strength and endurance adaptations. However, to determine the effectiveness of these exercises in actually conditioning the lumbar extensors over a chronic training period, and thus their value in prehabilitation, it is necessary to examine studies that have utilised a training intervention of exercises and performed testing that allows potential identification of adaptation in the lumbar extensors.

Chronic adaptation to specific exercise training for the lumbar extensors

Unfortunately, recommendations regarding effectiveness of a particular exercise on conditioning the lumbar extensors are often based solely on whether or not the muscle is active during a specific exercise.⁸⁵ Yet a number of training intervention studies have in fact examined the effect of exercises aimed at the lumbar extensors on strength and endurance. However, as noted, to determine whether any adaptation has potentially occurred in the lumbar extensors it is necessary to employ testing of lumbar extension strength or endurance in isolation (ILEX testing) or testing that allows differentiation between hip extension and lumbar extension (ie, EMG or imaging study as some have used^{34–37 86}).

Nicodemus⁸⁷ highlighted that although some improvement in ILEX strength resulted from roman chair TEX, this was

significantly less (~50%) than improvement produced by ILEX exercise. Verna *et al*⁸⁸ examined 8 weeks of training using roman chair TEX on the Biering-Sorenson test and ILEX strength. Their results indicated no effect of the training on ILEX strength. Mayer *et al*⁸⁹ reported similar results showing no effect of roman chair exercise on ILEX. This despite participants in the training group demonstrating an increase in endurance time during the Biering-Sorenson test.⁸⁸ These studies highlight the need for training interventions to confirm effectiveness and specificity of exercises for prehabilitation, in light of contradictory results from acute activation studies,⁵⁵ (ie, that acute lumbar extensor activity can sometimes be greater during TEX than ILEX). It seems the considerable degree of freedom to change kinematics⁶¹ and tendency for the lumbar extensors to derecruit^{45 46 61} may explain why studies do not show any adaptation resulting from TEX.^{88 89}

Although not all comparative trials, studies have offered insight into effectiveness of free weight exercise on the lumbar extensors. Pollock *et al*¹² reported that even healthy strength trained individuals who have engaged in such exercises (deadlifts, squats, good mornings, etc) still demonstrate some degree of lumbar extensor disuse atrophy suggesting that the exercises provide little stimulus to produce or maintain ILEX strength. Pollock *et al*¹² showed considerable increases in ILEX strength after ILEX training in healthy men previously engaged in heavy traditional resistance training for at least 1 year. They speculated that the magnitude of improvement (42–102%; considerably more than reported in studies of other exercises for other muscle groups; ~15–30%) were due to the lumbar extensors existing in a state of prior atrophy from chronic disuse as most exercises may not stimulate a conditioning effect. However this may also have been due to the highly motivated sample population.⁹⁰ A randomised controlled trial has recently been conducted⁹¹ comparing training using ILEX to the stiff-legged deadlift (which loads and significantly activates the lumbar extensors^{64 67}) upon ILEX strength and deadlift one repetition maximum (1RM). A significant increase in ILEX strength as well as deadlift 1RM was observed in the ILEX group after 10 weeks of training whereas the deadlift group only improved deadlift 1RM.

No studies examining the effects of floor and stability exercises on ILEX strength in asymptomatic populations were found through our literature search. Some have examined effects of floor and stability exercises on ILEX strength in symptomatic LBP participants.^{52 92} In addition, studies have examined the effects of such exercise on multifidus CSA in symptomatic LBP participants.^{34 93} However, it is unclear whether the results of these studies offer any insight as to the use of such exercises for prehabilitation means for asymptomatic participants. A study by Moffroid *et al*³⁷ has reported the effects of a series of floor-based specific exercises, including forms of TEX, on the Biering-Sorenson test and EMG fatigue parameters of the lumbar extensors at the level of L3 in asymptomatic women. They found participants increased endurance times in the Biering-Sorenson test after training but demonstrated no change in fatigability of the lumbar extensors.

Clearly there may be some potential for adaptation to occur in the lumbar extensors through these exercise approaches, though it seems support is equivocal and as such their value in prehabilitation may be questionable or at least requires additional investigation. It appears evidently difficult to stimulate improvement in the lumbar extensors through many supposedly 'specific' exercises. As mentioned, however, some machines offer means to isolate the lumbar spine providing ILEX and allow the lumbar extensors potentially to be primarily

responsible for work performed in isolation^{28 29} (ie, moving the resistance). However, a number of resistance machines aimed at the lumbar extensors exist commercially. Most of these machines do not employ the rigorous restraint system required for ILEX. Though many have been compared to ILEX in their ability to condition the lumbar extensors.

Results of studies that have compared the effect of commercial 'low back' machines with ILEX machines on ILEX strength measures are contrasting. Graves *et al*⁹⁴ compared training for 12 weeks using an ILEX device to more commercially available lower back machines, both which only restrain the legs. Their results showed, compared with controls, the ILEX group significantly increased ILEX strength, whereas no increase was seen in the other group despite an increased training load reported. Contrastingly, Nicodemus⁸⁷ reported no significant difference in ILEX strength increases when comparing ILEX with another low back extension machine. However, training was only conducted for a period of 5–6 weeks and also the methods including sample size are not reported. Mayer *et al*⁹⁵ sought to examine whether restraints required for ILEX are indeed required to condition the lumbar extensors. In this study a machine allowing ILEX through equipping of its restraints was used by both training groups to avoid the issue of learning effects confounding the earlier data where the ILEX groups tested as well as trained on the same machine.⁹⁴ Thus, both groups used the ILEX machine; one group with the restraints fastened to restrain the pelvis providing ILEX and the other seated in the machine without the restraints fastened properly. Their results suggested the restraint mechanism was not necessary to induce adaptation in the lumbar extensors; however, a concern with their results comes from the small sample size used.

Mayer *et al*⁹⁵ conducted a power analysis and used both Graves *et al*⁹⁴ and Pollock *et al*¹² to determine participant numbers for statistical significance at a power of $\beta=0.8$. However, combining the two studies likely reduced the calculated participant numbers required due to the effect size (ES) observed in Pollock *et al*¹² (ES=4.51; Calculated using equations from Whitley and Ball⁹⁶). The large treatment effect of this study¹² has been commented on previously by Graves *et al*⁹⁰ and noted that the unusually large improvements were due to the asymptomatic participants perhaps being experienced in resistance training and very highly motivated. No other study has demonstrated such large increases. Thus, it is likely that the results of Mayer *et al*⁹⁵ using a relatively small sample size ($n=18$) may represent a type II error (ie, failure to reject the null hypothesis). On the other hand, Graves *et al*⁹⁴ using a larger sample size ($n=77$) was likely more robust and less liable to statistical error. Indeed a recently published study⁵¹ that has looked at symptomatic participants employing the same method as the study of Mayer *et al*⁹⁵ using a greater number of participants ($n=42$) found that the restraint mechanisms used for providing ILEX are necessary to induce adaptation in ILEX strength. Nevertheless as with other studies of symptomatic LBP participants it is unclear to what extent the results of this study⁵¹ are applicable to the concern of prehabilitation.

Contrastingly however, a study by Parkkola *et al*⁹⁷ has demonstrated a significant increase in lumbar extensor CSA at the L4–L5 level through MRI as a result of TEX based exercise using commercial machines suggesting the possibility of some beneficial effect from such training. It is unclear whether differences in CSA adaptation may occur between approaches however as there is a lack of direct comparison studies.

These studies have examined adaptations in the lumbar extensors as a result of differing exercises. As strength and endurance

of the musculature may affect injury risk^{18–20} they provide insight to the potential value of these exercises for prehabilitation of the lumbar extensors in asymptomatic populations.

DISCUSSION

It has been conjectured for some time that when training using compound exercise (ie, TEX) the larger musculature (ie, hip extensors³⁰), take on greater proportional share of the load and this may affect adaptation produced.^{31–62} It appears the lumbar extensors are active during exercises such as roman chair or bench TEXs, free weight exercises such as the squat and deadlift, floor and stability ball exercise and extension-based resistance machines with the pelvis restrained (ILEX) and unrestrained (TEX). However, numerous chronic training studies examining exercises that involve TEX suggest that insufficient stimulus is provided to induce adaptation in the lumbar extensors. Various studies show that roman chair TEXs are significantly less effective compared with ILEX exercise,^{87–89} although some suggest adaptation might be possible.^{93–97} Free weight exercise (ie, squats, deadlifts) also appear unproductive.^{12–91} The majority of floor/stability ball-based exercise training studies have been conducted in symptomatic participants and contain numerous methodological issues rendering difficulty in drawing conclusion on their effectiveness.^{34–52–92} However, one study in asymptomatic individuals reports that fatigability of the lumbar extensors is not improved using this approach.³⁷ Finally, studies that have examined low back extension resistance machines appear to offer greater support for an approach that utilises restraint systems for ILEX.^{94–95}

There appears to be a distinct discrepancy between studies examining the acute activation of the lumbar extensor musculature and those examining chronic outcomes such as ILEX strength/endurance, change in CSA or change in fatigability. For example, approaches such as TEX, free weight exercise such as the deadlift, hip extension and TEX-based floor exercises and unrestrained seated machine extension exercises all demonstrate significant activation of the lumbar extensors which is normally considered as sufficient for inducing adaptation.^{81–84} The majority of chronic training studies, which have utilised ILEX strength testing in order to examine adaptation in the lumbar extensors, suggest however that these approaches do not effectively condition the lumbar extensors. For example, there is a drastic contrast between the activation levels shown in some studies for the deadlift (~75–125%^{65–67}) which does not appear to transfer to improved lumbar extensor strength when tested using an ILEX dynamometer after completion of a 10-week training programme utilising deadlifts.⁹¹ One possible explanation for this discrepancy shown in many studies might be due to the ‘specificity’ principle, relating to the exercise used during training and the ILEX testing mode used as an outcome.

In anticipation of this factor affecting measured outcomes, some authors have attempted to examine the purported superiority of ILEX for conditioning the lumbar extensors by utilising the same machine for exercise and testing outcomes with as well as without sufficient restraints during acute study^{27–28–76} and also during chronic training studies.^{51–95} These studies are contrasting in their findings as noted in the results section of this review with some showing greater activation and strength gains with ILEX,^{27–28–51} one that does not,⁹⁵ and another that found only a trend towards significantly greater results using ILEX.⁷⁶ As noted however, those studies suggesting ILEX may not necessarily have methodological limitations, such as smaller sample size, that are of concern. It should also be noted that one of the studies supporting ILEX utilised symptomatic participants⁵¹ and so again may not reflect adaptations in asymptomatic

participants. Yet, adding further light to the consideration of whether specificity is indeed an issue, an increase in ILEX strength from ILEX training is shown to also produce significant improvement in deadlift 1RM.⁹¹ Evidently, improved ILEX strength is associated with improvement in tasks involving TEX; however, performing these movements may not sufficiently stimulate the lumbar extensors themselves. Reyna *et al*⁹⁸ have also shown that maximum ILEX strength correlates well ($r=0.645$) with dynamic lifting tasks supporting this and highlighting that the lumbar extensors may be a weak link.²² Matheson *et al*⁹⁹ further reported that ILEX strength independently contributes to lifting capacity. Mooney *et al*¹⁰⁰ also reported significant improvement in lift capacity as a result of 8 weeks ILEX training in CLBP participants. That increased lumbar extension strength also improves strength in other compound TEX-based movements further reinforces its value in prehabilitation. This is apparently in contrast to what the ‘specificity’ principle would predict.

Some of the training studies examined utilised either EMG fatigue analysis or measurements of CSA of the lumbar extensors.^{37–97} In particular the study by Moffroid *et al*³⁷ demonstrated improved Biering-Sorenson test times in the absence of improvements in fatigability of the lumbar extensors. This might suggest that adaptation was limited to the hip extensors potentially due to the load sharing they engage in during TEX-based exercise.^{45–46} However, it is also possible that adaptation may have occurred in the thoracic extensors which have been shown to contribute to load sharing also during TEX exercise.⁶¹ Yet, the majority of chronic training studies have utilised some form of strength or endurance testing and the use of ILEX testing has been emphasised within this review as it allows isolation of lumbar extension movement from hip extension movement. However, it should be noted that the movement produced during ILEX is a product of the action of the lumbar extensor musculature (lumbar erector spinae and multifidus) as well as the thoracic portions of extensor musculature in addition to bilateral contraction of the quadratus lumborum. Thus, it is noted that although ILEX differentiates clearly between hip extension and lumbar extension, the lumbar extension movement produced is a result of the contribution of the lumbar and thoracic extensors. As such it might be deductively speculated that in those studies involving exercise allowing TEX movement and showing improvement in TEX based testing but not in ILEX testing, that adaptations may have occurred primarily in the hip extensors.^{88–91–94} An absence of improvement in ILEX would suggest that improvement in neither the lumbar or thoracic extensors has resulted. Contrastingly studies using ILEX exercise interventions do show improved ILEX strength⁹⁴ though it is more troublesome to discern whether the lumbar or thoracic extensors have improved in these instances. ILEX exercise activation studies do suggest that lumbar extensor involvement is greater than lower thoracic extensor^{28–29} and therefore that perhaps it is the lumbar extensors adapting to improve ILEX strength. Though in consideration of the discordance between other EMG studies and the chronic training studies examined this appears unsure. ‘Thoracic’ was not utilised as a term within our search strategy and so it is difficult to draw a fully informed conclusion on the topic of the thoracic extensors adaptations in response to any of the exercises reviewed.

Although evidence is suggestive that ILEX may be a more favourable approach to conditioning the lumbar extensors, future studies should endeavour to more rigorously compare the different exercise approaches noted here in comparison to ILEX exercise. Many have also examined exercise approaches but not incorporated testing that allows differentiation of the lumbar extensor

musculature. These future studies should employ valid, isolated testing of the lumbar extensors either through ILEX strength/endurance testing, EMG fatigue testing of the lumbar extensors, imaging study of adaptations to the lumbar extensors or ideally a combination of these methods. In addition, though there is evidence in occupational settings that prospective strengthening using ILEX reduces injury risk^{18 19} and there is reason to believe that conditioning the musculature reduces injury risk,²⁰ it would be of interest for future research to specifically examine this form of 'prehabilitation' in athletic populations.

It is beyond the scope of this review to suggest optimal means of manipulating variables (load, repetitions, repetition duration, volume, frequency) when employing exercises designed for the lumbar extensors. However, it would at present seem there is some evidence to support ILEX exercise to most effectively strengthen the lumbar extensors. Although loaded and active during any of the exercises reviewed, evidently the lumbar extensors and the movement they perform are limited when the pelvis is free to rotate through the action of the hip extensors. Although some have been previously put off by the potentially high cost of some ILEX devices there now exists a range of devices offering such exercise including inexpensive 'low tech' options (eg, Lumbar Extension Machine, MedX, Ocala, Florida; BackUp Dynamometer, Priority One Equipment, Grand Junction, Colorado; Lower Back Revival System, OriGENE Concepts BV, Delft, the Netherlands etc). The suggestion to utilise ILEX should not be interpreted as discouraging the use of TEX-based exercise as they may provide positive adaptations for the hip extensors^{37 88 91 94} (although again different exercises may offer different degrees of efficacy in this regard¹⁰¹), and may still have the potential for inducing adaptation of the lumbar extensors as some studies suggest (ie, TEX using benches or roman chairs,⁸⁷ hip extension and TEX-based floor exercise⁸⁶ and unrestrained machine based TEX^{95 97}). However, health and fitness providers and facilities, strength and conditioning coaches, athletes and the general population should consider the specificity of exercises if they have the goal of optimally conditioning the lumbar extensors specifically perhaps as a prehabilitation method for addressing low back injury and pain risk.

Major findings

- ▶ A variety of exercises have been proposed to effectively condition the lumbar extensors including benches and roman chair TEXs, free weights exercises (ie, deadlifts, squats, good-mornings, etc), floor and stability ball exercise (ie, TEXs, bridging, four-point kneeling, etc) and resistance machines (ie, those with and without pelvic restraints).
- ▶ All of these exercise approaches are evidenced to acutely activate the lumbar extensors, though to varying degrees.
- ▶ In contrast, chronic training studies suggest that, although many of these exercises may condition the hip extensors, most are less effective at conditioning the lumbar extensors.
- ▶ Exercise using pelvic restraints that allow ILEX appears potentially most effective at conditioning the lumbar extensors and may be of greatest value in prehabilitation.
- ▶ However, methodological concerns highlighted throughout this review emphasise the need for further chronic training studies utilising valid outcome measures to more firmly establish which exercise approaches are most effective at conditioning the lumbar extensors.

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REFERENCES

- 1 World Health Organisation. *The World Health Report 1998: Life in the 21st century: A vision for all*. Geneva: Office of Publications, World Health Organisation, 1998.
- 2 Office for National Statistics. *Social trends 30*. London: The Stationary Office, 2000.
- 3 Office for National Statistics. *Social trends 40*. Basingstoke: Palgrave Macmillan, 2010.
- 4 Waddell G, Burton AK. Occupational health guidelines for the management of low back pain at work: evidence review. *Occup Med (Lond)* 2000;51:126–35.
- 5 Walker BF. The prevalence of low back pain: a systematic review of the literature from 1966 to 1998. *J Spinal Disord* 2000;13:205–17.
- 6 National Institute for Health and Clinical Excellence. *Low back pain: early management of persistent non-specific low back pain*. London: Royal College of General Practitioners, 2009.
- 7 Graned H, Morelli B. Low back pain among retired wrestlers and heavyweight lifters. *Am J Sports Med* 1988;16:530–3.
- 8 Sward L, Hellstrom M, Jacobsson B, et al. Back pain and radiologic changes in the thoraco-lumbar spine of athletes. *Spine* 1990;15:124–9.
- 9 Kraft DE. Low back pain in the adolescent athlete. *Paediatr Clin North Am* 2002;49:643–53.
- 10 Bono CM. Low back pain in athletes. *J Bone Joint Surg Am* 2004;86:382–96.
- 11 Bahr R, Andersen SO, Loken S, et al. Low back pain among endurance athletes with and without specific back loading—a cross sectional survey of cross country skiers, rowers, orienteers, and nonathletic controls. *Spine* 2004;29:449–54.
- 12 Pollock ML, Leggett SH, Graves JE, et al. Effect of resistance training on lumbar extension strength. *Am J Sports Med* 1989;17:624–9.
- 13 Jones A. *The lumbar spine, the cervical spine and the knee*. Ocala, Florida: MedX Corporation, 1993.
- 14 Carpenter DM, Nelson BW. Low back strengthening for the prevention and treatment of low back pain. *Med Sci Sports Exerc* 1999;31:18–24.
- 15 Smith D, Bruce-Low S, Bissell G. Twenty years of specific, isolated lumbar extension research: a review. *J Orthop* 2008;5:e14.
- 16 DeLorme TL. Restoration of muscle power by heavy-resistance exercises. *J Bone Joint Surg Am* 1945;27:645.
- 17 DeLorme TL, Watkins AL. Technics of progressive resistance exercise. *Arch Phys Med Rehabil* 1948;29:263–73.
- 18 Mooney V, Kron M, Rummerfield P, et al. The effect of workplace based strengthening on low back injury rates: a case study in the strip mining industry. *J Occup Rehabil* 1995;5:157–67.
- 19 Matheson L, Mooney V. Employment screening and functional capacity evaluation. In: Liebensohn C, ed. *Rehabilitation of the spine: a practitioners manual*. New York: Lippincott Williams & Wilkins, 2006:276.
- 20 Stone MH. Muscle conditioning and muscle injuries. *Med Sci Sports Exerc* 1990;22:457–62.
- 21 Pollock ML, Graves JE, Carpenter DM, et al. Muscle. In: Hochschuler SH, Cotler HB, Guyer RD, eds. *Rehabilitation of the spine: science and practice*. St. Louis: Mosby, 1993:263–84.
- 22 Smidt G, Herring T, Amundsen L, et al. Assessment of abdominal and back extensor function. *Spine* 1983;8:211–19.
- 23 Petersen CM, Amundsen LR, Schendel MJ. Comparison of the effectiveness of two pelvic stabilization systems on pelvic movement during maximal isometric trunk extension and flexion muscle contractions. *Phys Ther* 1987;67:534–9.
- 24 Graves JE, Pollock ML, Carpenter DM, et al. Quantitative assessment of full range of motion isometric lumbar extension strength. *Spine* 1990;15:289–94.
- 25 Inanami H. *Iwai orthopedic hospital rehabilitation program*. Daytona: International Spinal Rehabilitation Update Symposium, 1991.
- 26 Graves JE, Fix CK, Pollock ML, et al. Comparison of two restraint systems for pelvic stabilisation during isometric lumbar extension strength testing. *J Orthop Sports Phys Ther* 1992;15:37–42.
- 27 San Juan JG, Yaggie JA, Levy S, et al. Effects of pelvic stabilisation on lumbar muscle activity during dynamic exercise. *J Strength Cond Res* 2005;19:77–81.
- 28 Da Silva RA, Lariviere C, Arsenault AB, et al. Pelvic stabilization and semisitting position increase the specificity of back exercises. *Med Sci Sports Exerc* 2009;41:435–43.
- 29 Lariviere C, Da Silva RA, Arsenault AB, et al. Specificity of a back muscle exercise machine in healthy and low back pain subjects. *Med Sci Sports Exerc* 2010;42:592–9.

- 30 Farfan HF. Muscular mechanism of the lumbar spine and the position of power and efficiency. *Orthop Clin North Am* 1975;6:135–44.
- 31 Dul J, Johnson GE, Shiavi R, et al. Muscular synergism II: minimum fatigue criterion for load sharing between synergistic muscles. *J Biomech* 1984;17:675–84.
- 32 Miltner O, Wirtz DC, Siebert CH. Lumbar extension exercise (MedX)—treatment for chronic low back pain—review and meta-analysis. *Z Orthop Ihre Grenzgeb* 2001;139:287–93.
- 33 Mayer J, Mooney V, Dagenais S. Evidence informed management of chronic low back pain with lumbar extensor strengthening exercises. *Spine J* 2008;8:96–113.
- 34 Hides JA, Richardson CA, Jull GA. Multifidus muscle recovery is not automatic after resolution of acute, first episode low back pain. *Spine* 1996;21:2763–9.
- 35 Mooney V, Gulick J, Perlman M, et al. Relationships between myoelectric activity, strength and MRI of lumbar extensor muscles in back pain patients and normal subjects. *J Spinal Disord* 1997;10:348–56.
- 36 Choi G, Raiturker PP, Kim M, et al. The effect of early isolated lumbar extension exercise program for patients with herniated disc undergoing lumbar discectomy. *Neurosurgery* 2005;57:764–72.
- 37 Moffroid MT, Haugh LD, Haig AJ, et al. Endurance training of trunk extensor muscles. *Phys Ther* 1993;73:10–17.
- 38 Stokes IAF, Henry SM, Single RM. Surface electrodes do not accurately record from lumbar multifidus muscles. *Clin Biomech(Bristol, Avon)* 2003;18:9–13.
- 39 Alaranta H, Hurri H, Heliovaara M, et al. Non-dynamometric trunk performance tests: reliability and normative data. *Scand J Rehabil Med* 1994;26:211–15.
- 40 Moffroid M, Reid S, Henry SM, et al. Some endurance measures in persons with chronic low back pain. *J Orthop Sports Phys Ther* 1994;20:81–7.
- 41 Mayer T, Gatchel R, Betancur J, et al. Trunk muscle endurance measurement: isometric contrasted to isokinetic testing in normal subjects. *Spine* 1999;20:920–7.
- 42 McGill SM, Childs A, Liebenson C. Endurance times for low back stabilization exercise: clinical targets for testing and training from a normal database. *Arch Phys Med Rehabil* 1999;80:941–4.
- 43 Latimer J, Maher CG, Refshauge K, et al. The reliability and validity of the Biering-Sorenson test in asymptomatic subjects and subjects reporting current or previous nonspecific low back pain. *Spine* 1999;24:2085–90.
- 44 Kankaanpää M, Laaksonen D, Taimela S, et al. Age, sex, and body mass index as determinants of back and hip extensor fatigue in the isometric Sorenson back endurance test. *Arch Phys Med Rehabil* 1998;79:1069–75.
- 45 Clark BC, Manini TM, Mayer JM, et al. Electromyographic activity of the lumbar and hip extensors during dynamic trunk extension exercise. *Arch Phys Med Rehabil* 2002;83:1547–52.
- 46 Clark BC, Manini TM, Ploutz-Snyder LL. Derecruitment of the lumbar musculature with fatiguing trunk extension exercise. *Spine* 2003;28:282–7.
- 47 Udermann BE, Mayer JM, Graves JE, et al. Quantitative assessment of lumbar paraspinal muscle endurance. *J Athl Train* 2003;38:259–62.
- 48 Hager SM, Udermann BE, Reineke DM, et al. Quantification of lumbar endurance on a backup lumbar extension dynamometer. *J Sports Sci Med* 2006;5:656–61.
- 49 Kankaanpää M, Taimela S, Laaksonen D, et al. Back and hip extensor fatigability in chronic low back pain patients and controls. *Arch Phys Med Rehabil* 1998;79:412–17.
- 50 Willemink MJ, Van Es HW, Helmhout PH, et al. The effects of dynamic isolated lumbar extensor training on lumbar multifidus functional cross-sectional area and functional status of patients with chronic nonspecific low back pain. *Spine* 2012;37:E1651–8.
- 51 Smith D, Bissell G, Bruce-Low S, et al. The effect of lumbar extension training with and without pelvic stabilization on lumbar strength and low back pain. *J Back Musculoskelet Rehabil* 2011;24:241–9.
- 52 Helmhout PH, Harts CC, Vlechtbauer W, et al. Isolated lumbar extensor strengthening versus regular physical therapy in an army working population with nonacute low back pain. *Arch Phys Med Rehabil* 2008;89:1675–85.
- 53 Helmhout PH, Harts CC, Staal JB, et al. Comparison of a high-intensity and a low-intensity lumbar extensor strengthening training program as minimal intervention treatment in low back pain: a randomized trial. *Eur Spine J* 2004;13:537–47.
- 54 Harts CC, Helmhout PH, De Bie RA, et al. A high-intensity lumbar extensor strengthening program is little better than a low-intensity program or a waiting list control group for chronic low back pain: a randomised clinical trial. *Aust J Physiother* 2008;54:23–31.
- 55 Kearns CF, Brechue WF, Bauer J, et al. Muscle activation during isolated and non-isolated lumbar extension exercises. *Med Sci Sports Exerc* 1997;29:S165.
- 56 Arokoski JPA, Kankaanpää M, Valta T, et al. Back and hip extensor muscle function during therapeutics exercises. *Arch Phys Med Rehabil* 1999;80:842–50.
- 57 Mayer JM, Graves JE, Robertson VL, et al. Electromyographic activity of the lumbar extensor muscles: effect of angle and hand position roman chair exercise. *Arch Phys Med Rehabil* 1999;80:751–5.
- 58 Mayer JM, Verna JL, Manini TM, et al. Electromyographic activity of the trunk extensor muscles: effect of varying hip position and lumbar posture during roman chair exercise. *Arch Phys Med Rehabil* 2002;83:1453–546.
- 59 Behm DG, Leonard AM, Young WB, et al. Trunk muscle electromyographic activity with unstable and unilateral exercises. *J Strength Cond Res* 2005;19:193–201.
- 60 Goncalves M, Barbosa FSS. Analysis of strength and resistance parameters of the lumbar spinae erector muscles during isometric exercise at different effort levels. *Rev Bras Med Esporte* 2005;11:108e–13e.
- 61 Lariviere C, Da Silva RA, Arsenault AB, et al. Specificity of a back muscle roman chair exercise in healthy and back pain subjects. *Med Sci Sports Exerc* 2011;43:157–64.
- 62 Pauly JE. An electromyographic analysis of certain movements and exercises. I. some deep muscles of the back. *Anat Rec* 1966;155:223–34.
- 63 Benson ME, Smith DR, Bybee RF. The muscle activation of the erector spinae during hyperextension with and without the pelvis restrained. *Phys Ther Sport* 2002;3:165–74.
- 64 Escamilla RF, Francisco AC, Kayes AV, et al. An electromyographic analysis of sumo and conventional style deadlifts. *Med Sci Sports Exerc* 2002;34:682–8.
- 65 Hamlyn N, Behm DG, Young WB. Trunk muscle activation during dynamic weight training exercises and isometric instability activities. *J Strength Cond Res* 2007;21:1108–12.
- 66 Chulvi-Medrano I, Garcia-Masso X, Colado JC, et al. Deadlift muscle force and activation under stable and unstable conditions. *J Strength Cond Res* 2010;24:2723–30.
- 67 Colado JC, Pablos C, Chulvi-Medrano I, et al. The progression of paraspinal muscle recruitment intensity in localised and global strength training exercises is not based on instability alone. *Arch Phys Med Rehabil* 2011;92:1875–83.
- 68 Shirado O, Ito T, Kaneda K, et al. Electromyographic analysis of four techniques for isometric trunk muscle exercises. *Arch Phys Med Rehabil* 1995;76:225–9.
- 69 Vezina MJ, Hubley-Kozey CL. Muscle activation in therapeutic exercise to improve trunk stability. *Arch Phys Med Rehabil* 2000;81:1370–9.
- 70 Souza GM, Baker LL, Powers CM. Electromyographic activity of selected trunk muscles during dynamic spine stabilization exercises. *Arch Phys Med Rehabil* 2001;82:1551–7.
- 71 Arokoski JP, Valta T, Airaksinen O, et al. Back and abdominal muscle function during stabilization exercises. *Arch Phys Med Rehabil* 2001;82:1089–98.
- 72 Arokoski JP, Valta T, Kankaanpää M, et al. Activation of lumbar paraspinal and abdominal muscles during therapeutic exercises in chronic low back pain patients. *Arch Phys Med Rehabil* 2004;85:823–32.
- 73 Mori A. Electromyographic activity of selected trunk muscles during stabilization exercises using a gym ball. *Electromyogr Clin Neurophysiol* 2004;4:57–64.
- 74 Stevens VK, Bouche KG, Mahieu NN, et al. Trunk muscle activity in healthy subjects during bridging stabilization exercises. *BMC Musculoskelet Disord* 2006;7:75.
- 75 Stevens VK, Vleeming A, Bouche KG, et al. Electromyographic activity of trunk and hip muscles during stabilization exercises in four-point kneeling in healthy volunteers. *Eur Spine J* 2007;16:711–18.
- 76 Udermann BE, Graves JE, Donelson RG, et al. Pelvic restraint effect on lumbar gluteal and hamstring muscle electromyographic activation. *Arch Phys Med Rehabil* 1999;80:428–31.
- 77 Walsworth M. Lumbar paraspinal electromyographic activity during trunk exercises on two types of exercise machines. *Electromyogr Clin Neurophysiol* 2004;44:201–7.
- 78 Willard FH. The muscular, ligamentous and neural structure of the low back and its relation to back pain. In: Vleeming A, Mooney V, Stoecart R, eds. *Movement, stability and low back pain: integration of research and therapy*. Edinburgh: Churchill Livingstone, 2007:85–94.
- 79 McGill S, Jucker D, Kropf P. Appropriately places surface EMG electrodes reflect deep muscle activity (psoas, quadratus lumborum, abdominal wall) in the lumbar spine. *Spine* 1996;29:1503–7.
- 80 Roy SH, De Luca CJ, Emley M, et al. Spectral electromyographic assessment of back muscles in patients with low back pain undergoing rehabilitation. *Spine* 1995;20:38–48.
- 81 Rodney KJ, Herbert RD, Balnave RJ. Fatigue contributes to the strength training stimulus. *Med Sci Sports Exerc* 1994;26:1160–4.
- 82 Schott J, McCully K, Rutherford OM. The role of metabolites in strength training. Short versus long isometric contractions. *Eur J Appl Physiol* 1995;71:337–41.
- 83 Drinkwater EJ, Lawton TW, Lindsell RP, et al. Training leading to repetition failure enhances bench press strength increases in elite junior athletes. *J Strength Cond Res* 2005;19:382–8.
- 84 Carpinelli RN. The size principle and a critical analysis of the unsubstantiated heavier-is-better recommendation for resistance training. *J Exerc Sci Fit* 2008;6:67–86.
- 85 Hamill J, Kuntzen KM. *Biomechanical basis of human movement*. 3rd edn. Philadelphia: Lippincott Williams & Wilkins, 2009.
- 86 Kamaz M, Kiresi D, Oguz H, et al. CT measurement of trunk muscle areas in patients with chronic low back pain. *Diagn Interv Radiol* 2007;13:144–8.
- 87 Nicodemus CL. Biomechanical study on the efficacy of back strengthening methods and machines. *Proceedings of the American Back Society*, 8–11 December 1999, Las Vegas, Nevada.

- 88 Verna JL, Mayer JM, Mooney V, *et al.* Back extension endurance and strength: the effect of variable-angle roman chair exercise training. *Spine* 2002;27:1772–7.
- 89 Mayer JM, Udermann BE, Graves JE, *et al.* Effect of roman chair exercise training on the development of lumbar extension strength. *J Strength Cond Res* 2003;17:356–61.
- 90 Graves JE, Pollock ML, Foster D, *et al.* Effect of training frequency and specificity on isometric lumbar extension strength. *Spine* 1990;15:504–9.
- 91 Fisher J, Bruce-Low S, Smith D. A randomized trial to consider the effect of Romanian deadlift exercise on the development of lumbar extension strength. *Phys Ther Sport* 2012;14:139–45.
- 92 Udermann BE, Mayer JM, Donelson RG, *et al.* Combining lumbar extension training with McKenzie therapy: effects on pain, disability, and psychosocial functioning in chronic low back pain patients. *Gundersen Lutheran Med J* 2004;3:7–12.
- 93 Danneels LA, Vanderstraeten GG, Cambier DC, *et al.* Effects of three different training modalities on the cross sectional area of the lumbar multifidus muscle in patients with chronic low back pain. *Br J Sports Med* 2001;35:186–91.
- 94 Graves JE, Webb DC, Pollock ML, *et al.* Pelvic stabilization during resistance training: Its effect on the development of lumbar extension strength. *Arch Phys Med Rehabil* 1994;75:210–15.
- 95 Mayer JM, Graves JE, Udermann BE, *et al.* Development of lumbar extension strength: effect of pelvic stabilization during resistance training. *J Back Musculoskeletal Rehabil* 2002;16:25–31.
- 96 Whitley E, Ball J. Statistics review 4: sample size calculations. *Crit Care* 2002;6:335–41.
- 97 Parkkola R, Kujala U, Rytokoski U. Response of the trunk muscles to training assessed by magnetic resonance imaging and muscle strength. *Eur J Appl Physiol Occup Physiol* 1992;65:383–7.
- 98 Reyna JR, Leggett SH, Kenney K, *et al.* The effect of lumbar belts on isolated lumbar muscle. Strength and dynamic capacity. *Spine* 1995;20:68–73.
- 99 Matheson L, Leggett S, Mooney V, *et al.* Contributions of aerobic fitness and back strength to lift capacity. *Spine* 2002;27:1208–12.
- 100 Mooney V, Matheson L, Holmes D, *et al.* Effect of focused strength training after low back injury. *North American Spine Society Annual Meeting*. San Diego, California, 1993.
- 101 Contreras BM, Cronin KB, Schoenfeld BJ, *et al.* Are all hip extension exercises created equal? *Strength Cond J* 2013;35:17–22.